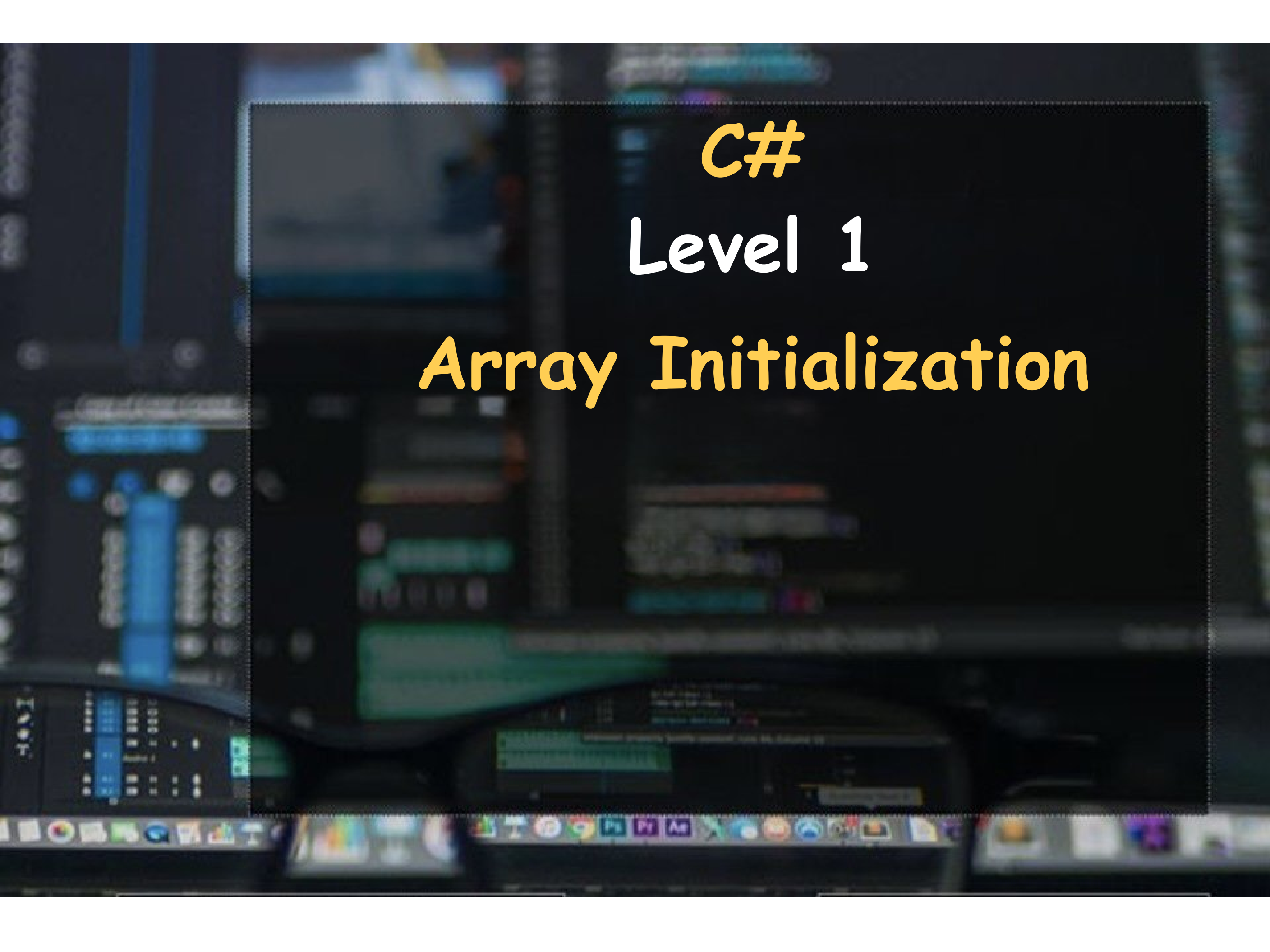


The background is a blurred image of a computer monitor. It shows a dark-themed code editor with various lines of code in different colors (blue, green, yellow). On the left side, there's a sidebar with some icons and text. At the bottom, a Windows taskbar is visible with several application icons like Chrome, Photoshop, and others.

C#

Level 1

Array Initialization



C#

Level 1

Array Initialization

Definition

Array Initialization

-  Giving values to the elements of an array.

-  Values can be assigned during declaration or later using indexes.

Explanation

Initialize During Declaration

-  Values can be provided directly inside { }.

-  C# automatically determines the array length from the number of values.

💡 Example

◆ `int[] numbers = { 1, 2, 3, 4, 5 };`

▣ Initialize Using Indexes

▨ An array can be created first and values assigned individually.

▨ Each value is assigned using `arrayName[index]`.

◆ `int[] age = new int[5];`

`age[0] = 12;`

`age[1] = 4;`

`age[2] = 5;`

`age[3] = 20;`

`age[4] = 30;`

▣ Array Index

▨ The first element has index 0.

▨ The last index is `Length - 1`.

◆ `int[] numbers = { 1, 2, 3, 4, 5 };`

`numbers.Length → 5`

`numbers[4] → 5`

Note

- ◆ Length represents the number of elements, not the last index.
- ◆ If Length = 5, valid indexes are 0 → 4.
- ◆ Accessing an invalid index throws `IndexOutOfRangeException`.



Comparison

▨ During Declaration

- ✓ `int[] numbers = { 1, 2, 3, 4, 5 };`
- ✓ `C#` determines the length automatically.

▤ Using Indexes

- ✓ `int[] numbers = new int[5];`
- ✓ Values are assigned individually using their indexes.

✦ Summary

- ✓ Arrays can be initialized during declaration or by assigning values through indexes.
- ✓ Indexes start from 0.
- ✓ Last index = Length - 1.
- ✓ Length = number of elements.

Thank you 😊